

Perspective

Design considerations for next-generation sorbent-based atmospheric water-harvesting devices

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THE BIGGER PICTURE Addressing the escalating global demand for clean and potable water necessitates innovative solutions, among which sorbent-based atmospheric water harvesting (SAWH) holds great promise. To unlock its full potential, significant advancements in sorbent- and device-level design are imperative. While prior work has focused on sorbent-level innovation, several challenges lie in scaling up to meet larger demands in smaller form factors. The complexity of system integration poses inherent challenges that warrant thorough exploration. While prior efforts have predominantly focused on material innovation for laboratory-scale applications, the challenging task of large-scale water production hinges primarily on the optimization of device-level design in various environmental contexts.

In this perspective, we present comprehensive design considerations for next-generation SAWH devices, including key components like sorbent selection, airflow, heating sources, and control. We aim to propel SAWH technology toward practical large-scale water production, addressing the pressing challenges of global water scarcity.

SUMMARY

To meet the increasing global demand for clean water, sorbent-based atmospheric water harvesting (SAWH) is a promising solution. Improving sorbent and device design is essential to enable substantial enhancement of the system's performance, but challenges remain, owing to the complexity of system integration. Prior work has largely focused on development of sorbent materials deployed in laboratory-scale systems to produce small amounts of water. However, the challenge of large-scale water production remains, falling largely on device-level design and optimization to efficiently utilize sorbent materials in a variety of environments. Here, we present comprehensive design considerations for next-generation SAWH devices, focusing on key components influencing system performance metrics. Various aspects of device design are considered, including sorbent selection and integration, airflow, heating sources, and condensation stages. In addition, we propose several figures of merit to unify sorbent and device development and provide a framework for objectively evaluating the performance of next-generation SAWH devices.

INTRODUCTION

Two-thirds of the global population, mostly located in developing countries, currently struggle with procuring potable water for at least one month out of the year.¹ Water scarcity leads directly to dangerous societal effects such as higher infant mortality and hospitalization rates,² food shortages,³ and disruptions to socioeconomic development.⁴ As climate change increasingly disrupts ecosystems and industrial chemicals continue to have an impact on freshwater resources, drinking water will continue its trend toward becoming a valuable and potentially rare commodity.⁵ In an effort to mitigate the growing need for drinking water globally, recent technological development has focused on sustainable sources of potable water generation, including

seawater desalination,^{6,7} dewing,^{8,9} fog harvesting,^{10,11} and, more recently, sorbent-based atmospheric water harvesting (SAWH).^{12–17} In this perspective, we discuss progress in, challenges to, and future directions for key components and system-level integration and optimization of SAWH devices. SAWH operates by using a sorbent, which is a class of material thermodynamically favorable under certain conditions to adsorb water vapor in proportionally large quantities relative to its mass. Sorbent behavior is temperature dependent in SAWH devices^{18–20}; sorbents desorb the water vapor adsorbate after thermal energy is put into the system, as either low-grade thermal waste heat or solar energy, among other heating source options. Total uptake and kinetics of the sorbent material have been identified as the two material attributes that govern the performance



Table 1. SAWH devices

	Passively driven devices		↔	Actively driven devices	
	LaPotin et al. ¹⁴	Park et al. ¹²		Hanikel et al. ³²	Shan et al. ¹⁵
Main energy source	solar (thermal)	solar (thermal)	solar (thermal and electric)	solar (electric)	grid electricity
Daily cycle number	single cycle	single cycle, partial steps	single cycle	multiple cycles	single cycle, partial steps
Adsorbent stage number	dual stage	single stage	four single stages	single stage	single stage
Adsorbent type	zeolite AQSOA Z01	PAM/Alg-CNF-CaCl ₂ hydrogel	LiCl@rGO-SA	MOF-303	LiCl-impregnated carbon felts
Condensation mechanism	subambient (fans)	ambient (natural convection)	subambient (fans)	subambient (compressor)	ambient (natural convection)
H ₂ O per day (L _{water} /L _{device})	0.01	N/A	0.009	0.003	0.056
H ₂ O per day (kg _{water} /kg _{device})	0.038	N/A	0.016	<0.010	0.097
User input between cycles	manual	manual	manual	partially autonomous	manual

Selection of current state-of-the-art SAWH devices, highlighting variability in subcomponent design and operation. The subscripts “water” and “device” refer to liquid water and SAWH device, respectively.

of SAWH devices, where uptake is defined as mass of water sorbed per mass of dry material and kinetics quantifies the rate of time at which the sorption occurs. Therefore, significant research has been done in optimizing these metrics for materials such as zeolites,²¹ silica gels,²² metal-organic frameworks (MOFs),^{23–25} and hydrogels.^{26–30} While previous technologies have been limited to areas with high humidity (above 60%) or required significant energy input to operate, SAWH has the potential to be a scalable, solar-driven system for supplying water to communities and to operate efficiently in a wide range of climates.³¹ Alternatively, SAWH systems could also be designed as highly portable and compact devices for individual users. Overall, it has been recently estimated that SAWH has the potential to provide sufficient drinking water for up to a billion people currently facing water scarcity.³¹

On a systems level, current SAWH devices fall into two broad categories based on the energy source required for sorbent regeneration: passively regenerating devices, which operate on a daily cycle with only solar energy input, and actively regenerating devices, which operate multiple cycles per day by employing electric or combustion (chemical) energy sources to initiate desorption. While both categories aim to maximize water output, each tackles different design constraints and targets fundamentally distinct applications. Passive systems are designed for maximum conservation of the limited solar energy input and are often intended, although not required, to be used as stationary devices in regions where electricity and potable water are rare commodities. In contrast, active systems often feature high-energy-density sources powering multiple cycles per day to maximize water output independent of the day-night cycle. Conventionally, active devices have the potential to be more spatially compact than their passive counterparts for the same daily water output, given the multicycle capability of the systems, which leads to the possibility of portable SAWH devices. Passive

devices thus far rely on solar thermal energy as the heating input for desorption,^{12,14} while a number of active systems instead use electricity either from photovoltaic (PV) panels or from the grid as the heating source.^{15,32} While numerous systems have been demonstrated to date, a subset of current state-of-the-art passive and active devices is shown in Table 1 to emphasize the variability in design constraints, such as energy source, cyclability, condensation type, water output relative to size, and relative degree of autonomy. There exists an emerging third class of device designs, which are considered either discretized continuous systems or batch-process passive systems, generating water similar to passive systems but integrating a batch process to achieve multiple cycles a day per system. An example of such a design is reported by Xu et al.,¹⁷ whose design features four independent sorbent beds whereby one bed is desorbing via solar thermal input while three beds are adsorbing at a given time. These beds are then manually swapped to continue the batch process while solar input is available.

To date, most literature on SAWH has limited the problem definitions and potential solutions all within the material level design space. However, integration of novel materials into a functional device often poses roadblocks for researchers, as materials rarely perform near their fundamental limit due to system non-idealities such as airflow control, vapor confinement, and thermal losses,¹⁴ as well as differences between material testing and device environmental conditions. Therefore, the importance of device-level design optimization has recently been discussed as essential to leverage the ideal performance of sorbent materials.^{33,34} Furthermore, this work demonstrates the potential for material uptake and kinetics to be a system-level, not a material-level, variable of interest, and thus it must be considered as a dependent variable for device optimization. When integrated into a device, a sorbent material may allow for beneficial attributes, such as compactness or energy efficiency, or inhibit

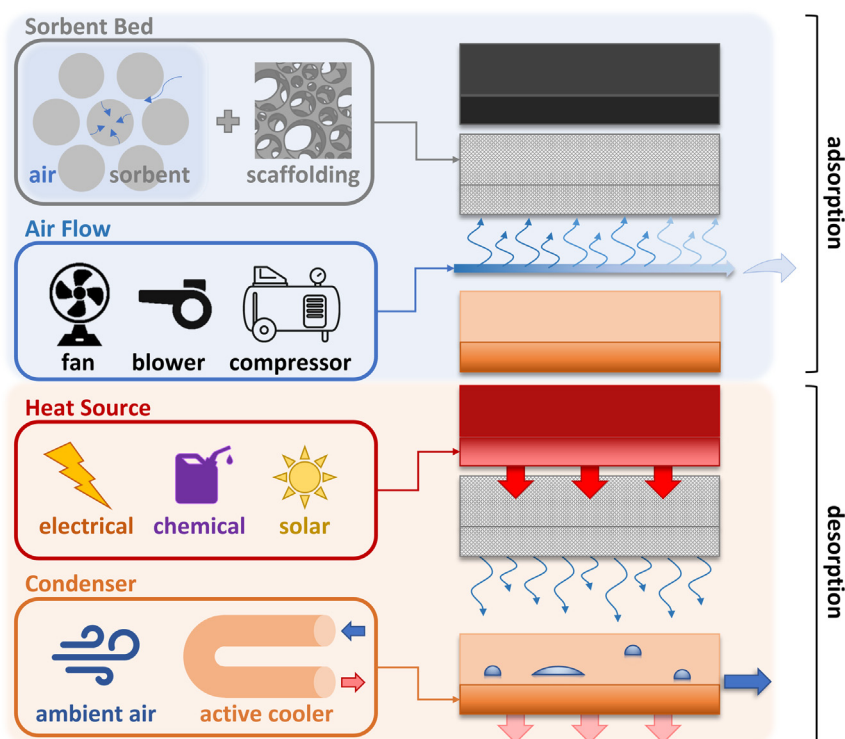


Figure 1. Assembly and operation of a SAWH device

SAWH devices cycle through adsorption and desorption phases. Blue arrows detail air/vapor transport during both adsorption and desorption, while red arrows illustrate heat flow through the device. Four critical device subcomponents are provided (sorbent bed, airflow, heat source, and condenser), with various options for executing the operation of each depicted.

posed into the sorbent bed, airflow mechanism, heat source, and condenser. During the adsorption phase, a sorbent that is connected/joined to a larger sorbent bed structure takes up water vapor from the ambient air, which is transported naturally or forced through the device. During the desorption phase, the bed is heated using one of many potential heat sources and water is desorbed from the sorbent, collecting on the actively or passively cooled condenser unit.

An essential aspect of SAWH devices is the sorbent choice, which drives the initial design and testing. Sorbent materials behave different from one another based

performance due to low uptake or slow kinetics given a supplied airflow. In this point of view, the system as a whole, not simply the sorbent material, must be evaluated and optimized with respect to critical subcomponent design parameters. Identifying these key design parameters remains an ongoing pursuit in the field of SAWH, which we aim to summarize in this work. To initiate the design of a SAWH device, three main heat and mass transport phenomena should be taken into consideration: (1) adsorption (i.e., airflow, fans, and sorbent selection), (2) desorption (heat source selection and sorbent thermal compatibility), and (3) condensation (condenser design and surface modifications). In this perspective, we aim to evaluate and optimize the system as a whole, not simply the sorbent material, by translating these main heat and mass transport phenomena into impacts and design constraints on the physical device subcomponents. First, we discuss the working principles and subcomponents of current state-of-the-art SAWH devices. Then, we identify the key design parameters that should be considered when designing a SAWH device for optimal adsorption, desorption, and condensation performance, with respect to system subcomponents. Finally, we provide unifying figures of merit that aim to assess the design and performance of SAWH devices on the material and system levels.

WORKING PRINCIPLES AND REQUIREMENTS OF WATER-HARVESTING DEVICES

Active and passive SAWH devices follow similar frameworks for operation, summarized in Figure 1. Several key device parameters are identified here with respect to subcomponents, decom-

posed into the sorbent bed, airflow mechanism, heat source, and condenser. During the adsorption phase, a sorbent that is connected/joined to a larger sorbent bed structure takes up water vapor from the ambient air, which is transported naturally or forced through the device. During the desorption phase, the bed is heated using one of many potential heat sources and water is desorbed from the sorbent, collecting on the actively or passively cooled condenser unit.

An essential aspect of SAWH devices is the sorbent choice, which drives the initial design and testing. Sorbent materials behave different from one another based on the ambient relative humidity (RH), which varies widely based on geographical location³⁵ (Figure 2A). Sorbent material operating conditions are often divided into five key thermodynamic regimes^{18,36}: (1) sorption of water molecules from ambient air at a given range of RH, defined by environmental conditions; (2) sensible heating of sorbent to increase material temperature; (3) desorption of water molecules from sorbent due to elevated temperature and/or decreased pressure and thus decreased vapor pressure at the sorbent-air interface; (4) sensible cooling of the air and desorbed water mixture; and (5) condensation and collection of desorbed water by ejecting the latent heat of condensation of water to the ambient. These thermodynamic regimes can be further grouped into two sorbent-specific phases that have an impact on device design, illustrated in Figure 2B as adsorption at a given environmental condition and desorption in a closed system. From these conditions we observe several device-level variables of interest, including material uptake, as well as fixed environmental design parameters. Water sorption isotherms, quantifying uptake as a function of RH, demonstrate the capability of the sorbent material to adsorb water vapor from ambient air at a given temperature. RH is related to pressure and temperature by Equation 1:

$$RH = \frac{P_{vap}(T)}{P_{sat}(T)} \quad (\text{Equation 1})$$

where P_{vap} is the vapor pressure of water vapor at the sorbent-air interface and P_{sat} is the saturation pressure in the system, describing the vapor pressure at which water vapor would condense. SAWH material isotherms for MOF-303,³² zeolite

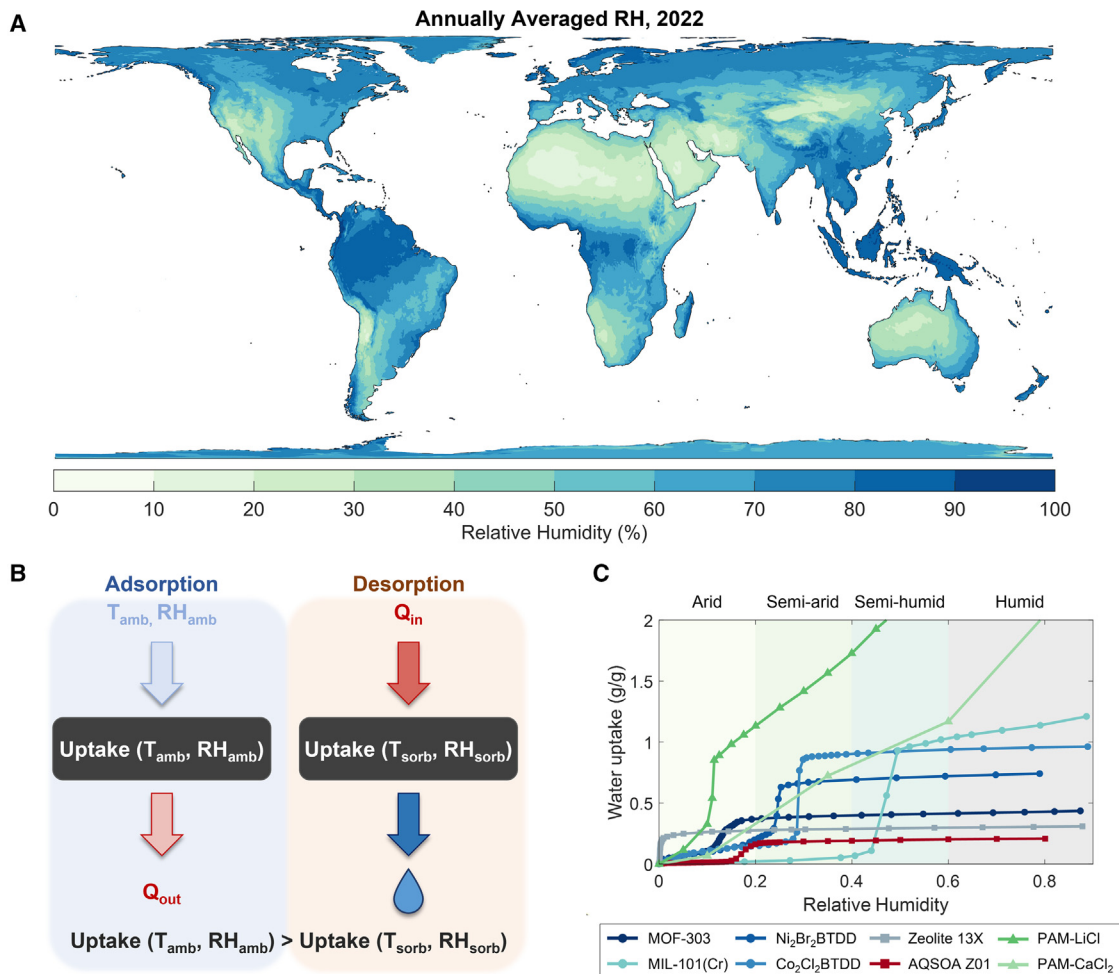


Figure 2. Sorbent considerations for SAWH devices

(A and B) (A) Average relative humidity of the earth in 2022³⁵ and (B) water flow (blue) and energy flow (red) for a sorbent and how each depends on RH.

(C) Practically, a SAWH device will utilize a sorbent whose isotherm demonstrates a steep change in uptake over the RH range in the environment in which it is deployed, to maximize the uptake differential between adsorption and desorption. Operationally ideal environments for specific sorbent materials are defined by the meteorologic climate zones arid, semi-arid, semi-humid, and humid.

13X,³² MIL-101(Cr),³⁷ Ni₂Br₂BTDD,³⁸ Co₂Cl₂BTDD,³⁹ AQSOA Z01,¹⁴ PAM-LiCl,⁴⁰ and PAM-CaCl₂⁴¹ are shown in Figure 2C, emphasizing the variability in material performance at a given environmental RH. Thus, application environment for device operation initially governs material choice, as shown in Figure 2, where a global average of RH dictates material choice to maximize uptake for a given location.

It is important to note that, from an energy point of view, dewing⁹ and SAWH become comparable for humid environments (RH ≈ 60%) at ambient temperatures ranging from 20°C to 30°C, and it becomes increasingly more energy intensive to use SAWH at higher humidities. SAWH is therefore ideal for low-humidity environments, identified as arid and semi-arid in Figure 2C. When choosing or designing a sorbent material, other factors that are often overlooked are adsorption enthalpy, density, and integration with device scaffolding. Adsorption enthalpy directly dictates how much energy must be supplied to the sorbent for desorption, influencing the heat source

design, while the material density and integration with device scaffolding affect device packaging, lifetime, and airflow requirements. It is therefore essential to design or choose a sorbent material based on not only uptake and rate of sorption for a given range of environmental humidities, but also available energy flux for desorption and integrability with the system. Once an initial sorbent material is chosen, engineers must consider the sorbent bed, airflow, heating source, and condenser, as well as device housing, sealing, and actuation to design, and optimize a SAWH system for application-specific performance targets. In the following discussion, we consider each subcomponent of a SAWH device separately, while highlighting parameters of interest that may have an impact on multiple subsystems and thus may be optimized during the design process. Preliminary work is done to relate several key parameters of SAWH to subcomponent design, with the aim to provide initial design considerations when pursuing next-generation SAWH devices.

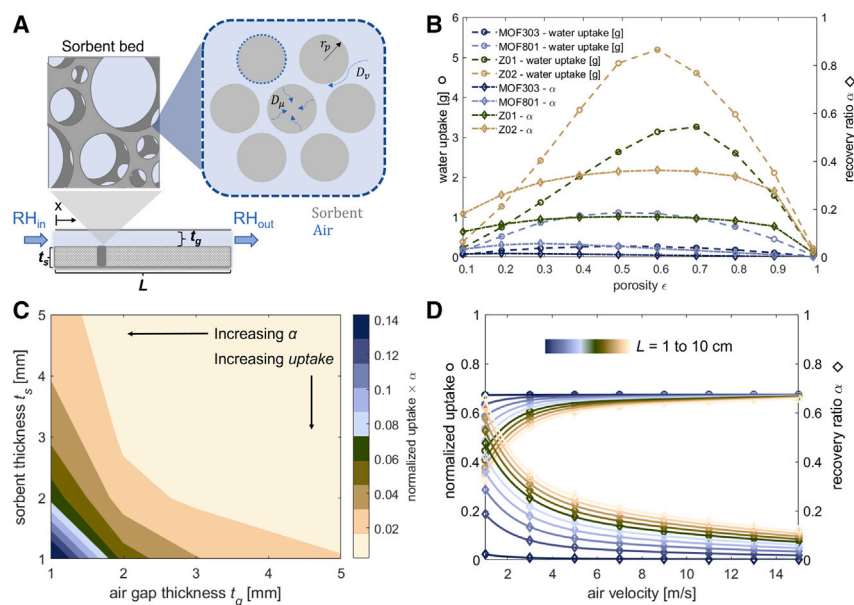


Figure 3. Impact of airflow in SAWH devices on recovery ratio and water uptake

(A) A schematic showing computational domain and optimization parameters (L , t_g , and t_s) and material properties for diffusivity.

(B) Uptake and vapor recovery ratio of various sorbent materials used in SAWH devices as a function of porosity, for a fixed sorbent bed geometry.

(C) AQSOA Z01 sorbent material normalized uptake times vapor recovery ratio (α) for varying air-gap (t_g) and sorbent (t_s) thicknesses, demonstrating that t_s drives uptake while t_g dictates α .

(D) Uptake and vapor recovery as a function of air velocity for a range of sorbent bed lengths.

Airflow

Directed airflow to and from the sorbent bed is a critical aspect of any SAWH device, as ambient air is the carrier of water vapor and affects both heat and mass transfer throughout the bed. While some single-cycle SAWH devices published to date may be able to rely on ambient wind to supply airflow to an exposed sorbent bed over the course of an entire night,^{14,15} high-water-output active devices increasingly rely on forced convection to shorten sorption times and thus increase daily water-harvesting cycles.^{17,32} One key metric for quantifying vapor delivery performance of the airflow is the vapor recovery ratio (α), defined as the percentage of water vapor that is adsorbed by the sorbent from the supplied airflow during the adsorption phase as a function of time (t). While generically a function of the flow's humidity ratio, the vapor recovery ratio for isothermal airflow can be calculated from the RH at the sorbent channel inlet (RH_{in}) and outlet (RH_{out}):

$$\alpha(t) = \frac{RH_{in} - RH_{out}(t)}{RH_{in}} \quad (\text{Equation 2})$$

Used as a metric for how efficiently water vapor can be captured by given sorbent bed geometry and ambient conditions, the vapor recovery ratio is limited by both material and device properties and depends on the ambient humidity. Using a simplified 2D approach to sorption and a heat transfer analogy for mass transfer, which sets the Sherwood number (Sh) as a constant for laminar flow, we can derive a scaling for RH in a flat sorbent bed for a given time:

$$RH(x) \sim \exp \left[-\frac{ShD_{air}}{u_{avg}t_g^2}x \right] \quad (\text{Equation 3})$$

where x is the position along the airflow path, t_g is the thickness of the vapor gap, u_{avg} is the average flow velocity of air through the bed, and D_{air} is the diffusivity of vapor in air. Equation 3 aids

us in determining parameters of interest for airflow optimization, which are expanded upon in Figure 3A. Another critical metric for quantifying sorption of water vapor from airflow is uptake, which for a SAWH device is both a spatially and a temporally dependent variable. To show how recovery ratio and uptake depend on both material

and system parameters, we employ a simplified schematic of airflow over a sorbent bed for a generic device. The physical picture is shown in Figure 3A, highlighting the key system design parameters of interest, including sorbent bed length (L), sorbent layer thickness (t_s), porosity (ϵ), airflow speed (u_{avg}), and air-gap thickness (t_g). Numerical simulation methodology is summarized in the supplemental information, modeling vapor flow as isothermal mass transfer to a 2D porous sorbent bed in COMSOL v.6.1 software. Heat generation due to enthalpy of adsorption is neglected for computational simplicity, due to negligible impact on both uptake and vapor recovery ratio given the thermal mass of materials and convective heat loss to the ambient. This assumption is further supported by previous SAWH devices, which feature sorbent beds with significant thermal mass that are cooled convectively, limiting the temperature variation of the subcomponent during adsorption.³²

On a material level, the uptake of the sorbent given local RH, determined from a material's isotherm, contributes to the vapor recovery ratio. For example, assuming an ambient RH of 60% and Ni_2Br_2BTDD as the sorbent material, which exhibits a step behavior in uptake at RH above 30% (Figure 2C), only up to half of the ambient water vapor can be captured by the sorbent, regardless of airflow rate or bed size. Once the local RH inside the sorbent bed at distance x from the entrance reaches 30% (Figure 3A), the sorbent material from this position onward no longer adsorbs further vapor from the airflow. This highlights the opportunity to enhance the vapor recovery ratio using materials with high water-uptake capabilities at low RH, including novel MOFs and salts and gel composites.⁴² Sorption kinetics, dictated by intercrystalline diffusivity (D_v) and intracrystalline diffusivity (D_μ), also affects water vapor transport in the sorbent and thus the vapor recovery rate. The system will end up with a small vapor recovery ratio if the airflow is too fast relative to the diffusion kinetics of the sorbent. Maximizing material kinetics will always yield a more efficient device; however, there exists an

optimum in terms of diffusive resistance given sorbent thickness, as a thinner sorbent will yield faster kinetics but will result in smaller total amount of water collected and/or larger bed size requirements. Thus, when designing a sorbent bed for SAWH, it is critical to consider the impact of sorbent bed length, width, air gap, and thickness on both the vapor recovery ratio and the fan power required for airflow.

Total water uptake and vapor recovery ratio of the sorbent bed depend strongly on sorbent bed porosity and sorbent material properties. Figure 3B shows results from the numerical COMSOL v.6.1 parametric study, highlighting water uptake after 30 min and initial vapor recovery as a function of sorbent bed porosity for several common sorbent materials recently published in AWH literature, including AQSOA Z01, AQSOA Z02,⁴³ MOF 303,³² and MOF 801.¹⁶ The same particle size of $r_p = 1 \mu\text{m}$ is applied to all materials, so that differences in results arise from material isotherms and intracrystalline diffusivity D_μ for a given porosity. Intracrystalline diffusion has been quantified in previous literature¹⁴ and varies based on material choice. Porosity of the sorbent bed is defined as:

$$\epsilon = \frac{V_{\text{air}}}{V_{\text{total}}} = 1 - \frac{\rho_b}{\rho_{\text{sorb}}} \quad (\text{Equation 4})$$

where ρ_{sorb} is the crystal density of the sorbent and ρ_b is the effective density of the sorbent bed. Powdered sorbents are often coated on a metallic sheet or foam for AWH applications; hence, the porosity is highly dependent on sorbent loading onto the scaffolding material, more so than packing of the crystal particles. A previously reported porosity of 0.67 is used when the porosity of the bed is fixed.¹⁴

System-level parameters also significantly affect both uptake and vapor recovery ratio metrics, regardless of sorbent material choice. For the purposes of a device-level airflow discussion, an AQSOA Z01 sorbent bed is evaluated, using a linear driving force model to evaluate mass transfer of water vapor to sorbent. Vapor diffusivity is assumed to be that of water vapor in ambient air, $D_{\text{air}} = 3.7 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$, and isothermal data on uptake versus RH is employed with respect to published data on AQSOA Z01 sorbent. All additional assumptions are listed in detail in the [supplemental information](#). This fixed sorbent bed is evaluated at a variety of lengths, airflow speeds, and thicknesses in Figures 3C and 3D to quantify performance dependence on system parameters. Figure 3C illustrates how total water uptake after a 30-min adsorption cycle is highly dependent on adsorbent layer thickness but exhibits negligible variation with changing air-gap thickness, as indicated by the increasing uptake arrow label. Conversely, initial vapor recovery ratio is dependent on the air-gap thickness, regardless of adsorbent bed thickness, for this system, as indicated by the increasing α arrow label. From this analysis, a small air gap and sorbent thickness is ideal for maximizing both uptake and vapor recovery ratio. In addition, as shown in Figure 3D, there is a direct trade-off between uptake and vapor recovery ratio with respect to airflow velocity. Faster airflow results in higher uptake values, but lower vapor recovery ratios. This trade-off demonstrates the need for tailoring system designs when considering different operational targets: if the

goal is maximum performance, airflow speed should be maximized; however, if energy efficiency is a restrictive design parameter, a high vapor recovery ratio should be targeted. A longer sorbent bed (in the direction of airflow) results in a higher vapor recovery ratio for a given airflow speed, but a lower uptake for the fixed amount of sorbent and thus inefficient use of the material.

By incorporating an ideal fan into this system analysis, we can describe the power required to flow a fixed mass flux of air through a system given bed geometry and sorbent material. Vapor recovery ratio, which depends on material choice and system design, directly represents the amount of water vapor available in the airflow and has the potential to vary fan power requirements. Figure 4 shows how vapor recovery ratio of the device strongly influences fan power requirements, particularly for lower RH environments, which often correlates to sites of severe water scarcity. Fan power is ideally generalized in Figure 4 as $\text{Power} = F * \Delta P$, where F is the volumetric flow rate of air over the sorbent bed and ΔP is the pressure drop across the sorbent bed, as estimated by the Darcy-Weisbach equation for pressure loss through parallel plates:

$$\Delta P = L \frac{96}{Re} \frac{\rho_{\text{air}} U_{\text{avg}}^2}{2D_h} \quad (\text{Equation 5})$$

where L is the length of the sorbent bed parallel to the direction of flow, U_{avg} is the cross-sectional average airflow speed, and Re is the Reynolds number of the flow calculated based on the hydraulic diameter $D_h = \frac{(2t_g w)}{t_g + w}$, where t_g is the gap between sorbent beds and w is the width of the bed into the page. A smooth pipe-flow analogy is utilized due to the dilute nature of water vapor in ambient air; however, this serves as only an approximation, as the sorbent bed features some degree of roughness and also slight mass transfer out of the bulk airflow due to water sorption. Typical sorbent bed designs aim to maximize the surface area of the sorbent and packing density of the bed, thus $t_g \ll w, L$ and a parallel plate assumption is valid. In this flow scheme, the results show that fan power requirements are dependent on both the vapor recovery ratio of the flow and the pressure drop through the bed (Figures 4A and 4B). Regardless of recovery ratio, power requirements decrease for both higher humidity inlet conditions and larger sorbent bed gaps per unit length, defined as $\frac{t_g}{L}$. Larger bed gaps t_g and shorter bed lengths L both reduce the pressure drop; however, the shorter channel length can lead to a lower α , regardless of material performance. An optimal bed geometry relative to fan power requirements can be obtained using this approach and quantified for a range of vapor recovery ratios, allowing for flexibility in modeling a system with a given sorbent uptake/kinetics. However, the challenge of achieving a target vapor recovery ratio remains, as bed geometry affects both vapor recovery ratio and fan power requirements. Airflow in AWH devices is a highly coupled design optimization problem, one future work must address to achieve situationally optimal water harvesting.

Heating sources

Heating is ubiquitous in SAWH systems, as all devices discussed here require thermal energy input for desorption.

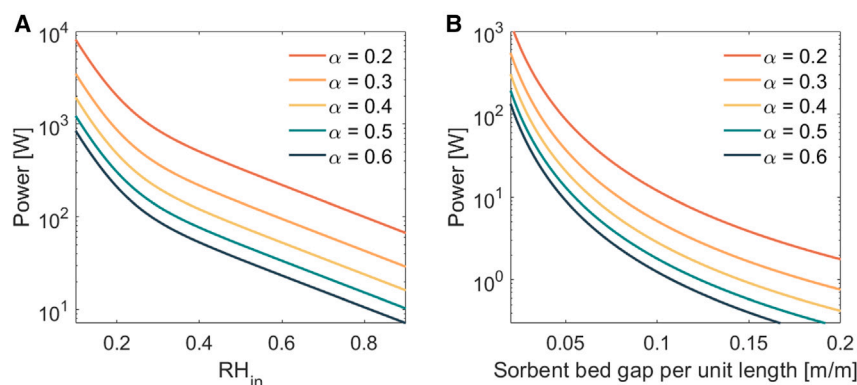


Figure 4. Impact of air flow in SAWH devices on fan power consumption

(A and B) The range of ideal power requirements for airflow across a sorbent stage, which achieves a given vapor recovery ratio α , with respect to (A) the relative humidity of the inlet air flow and (B) the air-gap thickness t_g per unit length L .

Figure 5 compares heat sources for SAWH devices based on a relative system, where a larger encompassed area equates to a higher score and thus a more desirable heat source based on the corresponding criteria. Scoring high on the carbon footprint category reflects a fuel source that has a higher carbon cost comparatively, and scoring high in the mass- and volumetric-density categories corresponds to a compact heat source on a kilojoule per kilogram and kilojoule per cubic meter basis, respectively. Complexity and adaptability are implemented as qualitative metrics, based on current heating systems utilizing each heat source, and discussed further in the next section. Furthermore, cost is also a critical metric when determining an appropriate heating source for practical AWH devices. In Figure 5, a higher score is associated with lower cost for each heating source, based on similar commercial components rather than practical device-integrated heating sources.

When evaluating thermal energy sources for AWH devices, it is crucial that a fair comparison is made between all options. Here, we discuss the efficiencies associated with each energy source as it pertains to delivering thermal energy to the sorbent bed for desorption, considering not only the device-level heating source but also any auxiliary components or infrastructure required to operate the heating component. For consistency, this discussion assumes room temperature environmental conditions; however, heating source operation and performance are also environmentally dependent, and thus operating conditions should be considered when selecting a specific SAWH device heating source.

As shown in Figure 5, electrical heating with resistive elements is a simple solution, which can be widely implemented, regardless of environmental conditions or bed design. Electrical heating is also much faster in terms of preheating relative to other thermal energy sources, which is necessary, as desorption is initiated only when a fixed temperature T_{des} is achieved (material dependent). Also, electrical heating is fairly energy dense, reducing ramp-up times for a multicycle device and allowing for precise temperature control. However, electrically heated sorbent beds suffer a large efficiency penalty upstream, with a typical diesel generator maintaining a fuel-to-electricity conversion of less than 30%.^{44,45} Furthermore, electrically heated devices require additional infrastructure, potentially including batteries, AC/DC converters, transformers, and other electronic

controllers/regulators. A battery-based heating system must also consider electrical charging time, which may negatively affect water harvesting cycle times. For a stationary AWH device with access to the electrical grid this may be a viable option for device operation, but the cost of electricity still limits the applicability of this approach.

Waste heat is an attractive strategy for either conductive or convective heating of the sorbent bed, as this thermal energy source comes at no cost to the user. Waste heat is simple to implement in an AWH device and has the potential for an efficiency above 90% associated with thermal energy delivery to the sorbent bed, given current heat-exchanger capabilities. Despite being an attractive option, waste heat is highly application dependent and lacks adaptability to different environments, operational situations, and water demands. It is also not an energy-dense source, as air is the thermal carrier, and low-grade waste heat poses a particular difficulty as the airflow temperature may be near the desorption temperature of the sorbent. A small temperature variation causes waste heat to be a slow heating source, with long ramp-up times, requiring a diligent thermal mass minimization of the sorbent bed and all other subcomponents along the thermal path. Insulation and mass minimization and conductivity maximization are crucial design parameters for an AWH device intending to operate on waste heat.

Flame or catalytic combustion is the most energy-dense thermal source option for SAWH devices and is an attractive contender for portable or compact designs.⁴⁶ Nevertheless, combustion heating is a complex approach that often requires a minimum ignition temperature, compressed air tanks or air entrainment, fuel/air manifold, mixing, and careful control of system thermal mass, conductivity, and flow regulation to maintain combustion while achieving the target sorbent desorption temperatures.⁴⁷

Solar thermal input to desorb in a SAWH device is currently the most-used heat source option, relying on selective solar absorber materials (SSAs) to capture over 95% of incident solar energy and convert it to thermal energy.⁴⁸ SSAs are low-cost materials relative to combustor, electrical, or waste heat components and provide thermal energy to the sorbent bed at no additional fuel cost. Solar thermal devices do not output any carbon emissions or require any additional infrastructure, making them ideal for off-grid, cheap, stationary applications. Yet, dependency on solar input for desorption makes SAWH devices of this architecture reliant on geographic location, weather, and time of day, resulting in an intermittent and comparatively less reliable and adaptable system.

From a fuel perspective, waste heat and solar energy are effectively free energy sources, while chemical heating is fairly

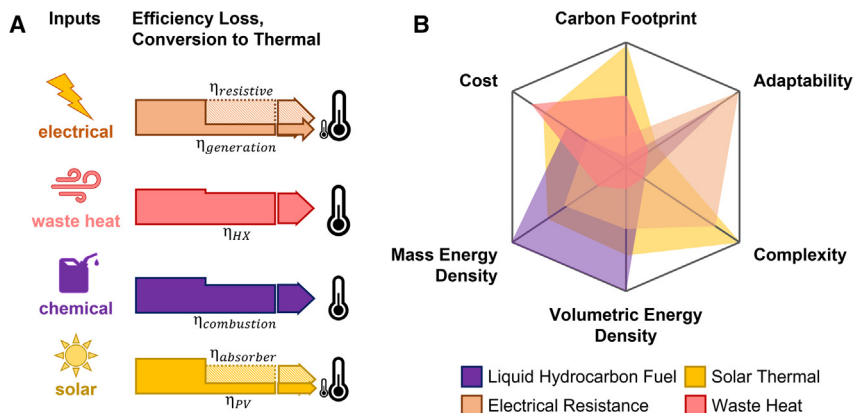


Figure 5. Heat source considerations for SAWH devices

(A and B) Summary of potential heating sources for SAWH devices emphasizing (A) the efficiency losses involved in converting the energy to heat, where some heating configurations lead to more (larger thermometer) or less (smaller thermometer) thermal energy relative to initial input potential, and (B) device-level pros and cons of specific examples from each input subset. A larger distance from the center equates to a better energy source relative to others for each metric, specified as a lower cost, higher energy and density, lower complexity, and higher adaptability.

cost effective compared with electrical input when fuel-to-electricity conversion efficiency is considered (Figure 5B). However, this cost hierarchy shifts substantially when considering integration of heat sources in devices. Electrical resistance heaters are common components and widely available in cost-competitive commercial markets. Flame and catalytic combustors are also commercially available, although the component complexity of these systems makes them more costly than electrical counterparts. A plethora of solar adsorbers exist and are often purchased for laboratory-scale SAWH devices but consist of optimized material chemistries to achieve high solar absorptivity and thus are more expensive than steel or copper electrical heaters. Finally, while no waste-heat-driven SAWH devices currently have been published, similar compact heat exchangers provide estimates for the cost of implementing such a thermal delivery system into a device. Various metal heat exchangers may be considered for waste heat delivery use, and while requiring more mass and volume than electrical heaters of the same material, do not require novel material development, similar to solar adsorbers and combustor heating units. Therefore, when considering the total cost of producing a SAWH device powered by each of the four heating sources considered in Figure 5, the following qualitative cost comparison arises: electrical is estimated as the cheapest system to produce, followed by solar thermal, then waste heat systems, and finally chemical combustor systems. However, it is important to note that, while the initial development of each system may differ in costs comparatively, fuel source costs throughout the lifetime of the device make solar and waste heat systems by far the lowest cost, as they require no capital for consumables such as fuel or electricity.

Similar to cost, carbon emissions from each potential heating source are an essential metric to consider when designing technology for the green market. While the manufacturing of components necessary for waste heat and solar thermal systems themselves have associated carbon costs, the carbon cost of continued operation is negligible compared with chemical heating systems such as flame or catalytic combustors, whose chemical-to-thermal reaction processes directly emit greenhouse gases to the environment. Electrical systems have a wide range of potential carbon footprints, ranging from low, if

powered by a renewable energy source such as wind or solar PV, to higher than chemical systems if powered by a diesel generator or fossil fuel equivalent.

Three pathways are available for thermal transport in a SAWH device: conduction, convection, and radiation. A device may heat the sorbent bed directly by either physical contact between a heat source and the sorbent bed or capturing incident solar radiation. Alternatively, a device design may indirectly heat the sorbent bed by first heating a sweep air, which then flows over the sorbent to initiate desorption. Radiation and conduction through the sorbent bed require engineers to consider the thermal mass of the system and temperature distribution across the bed, but does not require subcomponents in addition to the heat source. Alternatively, a convective desorption approach requires substantial infrastructure for sweep air control and propagation but has the potential to desorb quicker due to the higher vapor capacity of hot air versus cooler air. Air also has a lower thermal energy storage compared with metals, which can potentially contribute to a device with faster ramp-up and ramp-down times.

Condensers and water collection in SAWH

Once water vapor is released from a sorbent bed using thermal energy, SAWH devices rely on a condenser to collect the water. Previously, various condenser designs and configurations have been proposed. When solar energy is used as a heat source, transparent materials (e.g., glass, poly(methyl methacrylate) [PMMA]) were often vertically placed on a side wall⁴⁹ or inclined on the ceiling^{50,51} for solar energy to transmit through the housing. Similar to solar distillation systems, the heated water vapor can be desorbed from the sorbent bed, condensed at ambient temperature on the transparent material, and collected through a passive heat-sink process. Although an effective initial strategy, glasses or polymer materials with low thermal conductivity ($k < 1 \text{ W/(m}\cdot\text{K)}$) provide poor heat and mass transfer performance. Moreover, condensing droplets on the walls scatter the sunlight and degrade the overall water collection rate.

For the majority of SAWH devices, the condenser is a hydrophilic material such as glass or metal.^{13,52} On a wetting hydrophilic subcooled surface, condensing water vapor forms a condensate film, which deteriorates the heat transfer due to

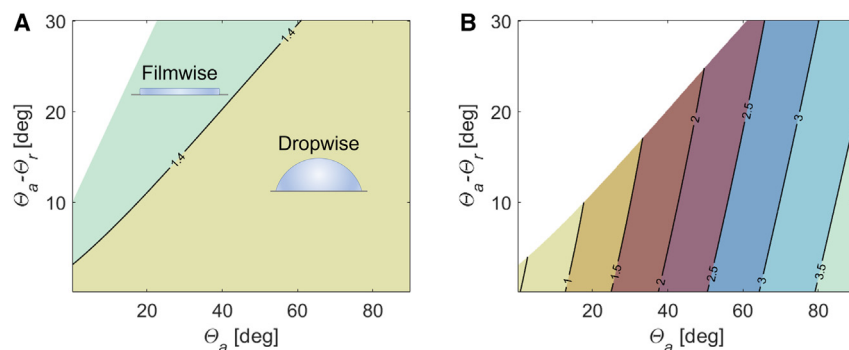


Figure 6. Condenser design for SAWH devices

(A) Condensation regime map as a function of advancing contact angle, θ_a , and contact angle hysteresis, $\Delta\theta = \theta_a - \theta_r$. The green and tan shaded regions represent filmwise and dropwise condensation, respectively.

(B) Droplet height map at departure during dropwise condensation as a function of advancing contact angle and contact angle hysteresis. The units are in millimeters (mm). As the advancing contact angle increases, the droplet height at departure increases.

the low thermal conductivity of water ($k \approx 0.6 \text{ W/(m}\cdot\text{K)}$). This phase-change process is commonly called filmwise condensation. In contrast, when vapor condenses on a non-wetting hydrophobic surface, small discrete droplets grow, coalesce, and shed via gravitational body forces. We refer to this behavior as dropwise condensation, and it typically yields increases in heat and mass transfer rate by an order of magnitude compared with the filmwise condensation.⁵³ More recently, combinations of surface micro-/nanostructures and hydrophobic coatings have enabled dynamic condensate droplet ejection,^{54,55} allowing further enhancement to heat transfer performance via coalescence-induced droplet jumping.⁵⁶ Although a variety of hydrophobic promoters have been developed over many decades,^{57–61} achieving long-term durability during steam condensation has been challenging when thin ($<1 \mu\text{m}$).^{62,63} In addition, the condensation heat transfer degrades significantly in the presence of the non-condensable gases (NCGs),⁶⁴ with an increasingly prominent effect on dropwise condensation.⁶⁵ Given that the air in the earth's atmosphere contains $>95\%$ NCGs by mole fraction on average, a recent study suggests that the filmwise condensation mode is a preferred mode of condensation for atmospheric water harvesting,⁶⁶ since vapor diffusion is the dominant mechanism of condensation.¹³

Despite the fact that filmwise condensation can potentially provide higher water collection rates, hydrophilic surfaces are prone to adsorbing atmospheric volatile organic compounds (VOCs) or aerosol particles when exposed to the atmospheric air in practical settings. Atmospheric VOCs can increase the hydrophobicity of the surface within a few days^{67–71} and significantly affect condensation behavior.^{72,73} Transition from filmwise to dropwise condensation typically occurs at the droplet Bond number $Bo \approx 1.4$ for droplets immediately before departure, where the droplet Bo is defined as $Bo = R^2/l^2$, where R is the characteristic length of the droplet, $l = \sqrt{\gamma/\rho g}$ is the capillary length, γ is the surface tension, ρ is the density, and g is the gravitational acceleration (Figure 6A).⁷⁴ When a hydrophilic surface is exposed to the ambient air and adsorbs VOCs, it easily becomes hydrophobic by increasing its advancing contact angle (θ_a) and undergoes dropwise condensation. However, VOC adsorption at the interface results in large contact-angle hysteresis ($\Delta\theta = \theta_a - \theta_r$, where θ_r is the receding contact angle), which leads to a significant amount of water retention on a condenser after each water-harvesting cycle.

When VOCs increase the advancing contact angle and its contact-angle hysteresis, the corresponding droplet height increases significantly (Figure 6B). For portable/compact AWH devices, spacing between the adsorbent bed and the condenser needs to be fairly close to minimize the volume of the system. If the adsorption bed-to-condenser spacing is too close, condensing droplets can touch the adsorption bed. While major progress has been focused on sorption materials development, condenser materials and designs have not received much attention for AWH devices. Recent analysis promotes the use of hydrophilic surfaces for filmwise condensation in AWH systems; however, difficulties associated with this condensation mechanism, including atmospheric contamination, water retention, and surface degradation, must be addressed to optimally integrate the condenser subcomponent into next-generation SAWH devices.

Actuation and autonomous operation

Current devices rely on manual operation, which may consist of researchers swapping sorbent beds into and out of the system or applying and removing covers/clamps/fixtures each sorption cycle. As AWH technology transitions to meet higher water output requirements and thus higher cyclability of a device, the need for automation becomes increasingly important. This demand is noted in a recent publication by Xu et al.,¹⁷ which features a design the authors say has the potential for automation. However, no truly autonomous SAWH device has been presented at the time of this publication. Whether the device targets large, household-to-building-scale water production or individual water production, the aim is to minimize user inputs to the system to promote usability and deployment of the device. There are several potential pathways for automation, including heat source automation, airflow automation, and mechanized transition from sealed to open device for desorption and adsorption, respectively. Control over the heat source flux allows the device to transition from adsorption to desorption, while airflow control minimizes power consumption by deploying the air delivery subcomponent only when necessary for adsorption, cooling, or desorption, depending on the device design. Both of these systems can be automated electronically but require sensory feedback to operate in a closed-loop configuration. For heating, this can simply be thermocouples strategically placed in the device, whereas for airflow, a humidity sensor has the potential to

provide feedback regarding the current environment such that the system can adjust flow and heating accordingly.⁷⁵

Water vapor retention and condensation during the desorption stage are highly reliant on sealing of the device, as even a small rate of vapor leakage can be compounded into a significant efficiency loss over the course of a desorption cycle. Contrarily, the device must be exposed to ambient air during the adsorption stage. Actuation of seals is thus essential for device performance. Both static and dynamic sealing are mature subjects in several engineering fields, including industrial turbines⁷⁶ and thermodynamic cycles.⁷⁷ While accepted standards for sealing designs, clamping pressures, and gasket sizes are frequently deployed for pumping and directing fluids,⁷⁸ AWH devices have yet to implement optimized sealing solutions. Dynamic sealing, in particular, poses several challenges to the next-generation AWH devices, as dynamic seals must be optimized to minimize both frictional wear and leakage, which are often mutually exclusive tribological objectives.⁷⁹ When mechanizing a SAWH device, it is important to minimize moving interfaces in the sorbent bed. By minimizing these sealable surfaces, we seek to reduce the risk of leakage during desorption and improve thermodynamic efficiency overall. Transitioning from user input of static seals to automation of dynamic sealing interfaces poses a significant design challenge in the field of SAWH devices, and future development must prioritize the sealing mechanisms to produce reliable and high-performance water-harvesting systems.

PERFORMANCE METRICS FOR SAWH

Current performance metrics are not standardized and thus tend to show favorable operation given a narrow environmental scope or application space. For example, passively driven solar devices cite specific water production (SWP) as a key metric for minimization, defined as liters of water output from the device daily per solar footprint, whether that be an adsorber or a solar panel. While helpful to quantify the performance of the sorbent stage given a specific environmental condition and a solar energy source, it does not include the impact of all the other system components, such as condenser units, power banks, and sealing/housing fixtures. Furthermore, isolating the volumetric scaling to a solar footprint allows for a high degree of variability between cited areas of PV panels or solar adsorber material and the total volume of the device. Specific energy consumption (SEC) is another metric of recent interest, and while inversely proportional to energy-to-water efficiency η_{thermal} , SEC couples within the additional impact of water loss due to device leakage and condensation inefficiencies, which is not reflected in an idealistic thermodynamic efficiency η_{thermal} . SEC and thermal efficiency metrics capture within each the effectiveness of energy conversion, vapor condensation, component energy storage, and both thermal and mass leakages on the system level. However, many emerging application spaces for SAWH are less concerned with energy efficiency and instead target compact systems that maximize the performance of a given material. Here, we propose a set of performance metrics that decompose device targets into SEC, sorbent layer packaging (SLP), thermal energy efficiency of the system (η_{thermal}), and water-harvesting efficiency of the system (η_{AWH}). These four terms not only reflect

conventional SAWH comparison metrics, such as SEC and thermal efficiency, but also standardize the quantification of future SAWH devices with respect to system level targets: size of the system (SLP) and performance of the device regardless of material choice (η_{AWH}). Equations 6, 7, 8, and 9 describe each metric in terms of quantifiable device variables:

$$SEC = \frac{Q_{in}}{V_{\text{water}}} \quad (\text{Equation 6})$$

$$\eta_{\text{thermal}} = \frac{m_w h_{fg}}{Q_{in}} \quad (\text{Equation 7})$$

$$SLP = \frac{V_{\text{device}}}{V_{\text{sorb}}} \quad (\text{Equation 8})$$

$$\eta_{\text{AWH}} = \frac{\frac{V_{\text{water}}}{\text{day}}_{\text{actual}}}{\frac{m_{\text{sorb}} \text{uptake}_{\text{swing}} N_{\text{cycles}}}{\text{day}}} \quad (\text{Equation 9})$$

Due to the low-density nature of sorbent materials and the large volume of air space contained within SAWH devices, volume rather than mass is the biggest constraint to portable/compact SAWH design.³³ However, current performance metrics for SAWH neglect volume in the evaluation of a device design, and variations in reporting device size make it difficult to fairly compare between technologies. As proposed, the SLP metric quantifies the use of device volume with respect to the minimum size of an ideal device, which would contain only the sorbent bed. Therefore, this value trends similar to SEC, where a lower value reflects a more efficient use of space for a given set of potential designs. It is essential when evaluating a design's SLP that all components required for producing liquid water from ambient air are included in the total device volume, including power units, condensers, and all external housing. Furthermore, we standardize the volume of the sorbent bed as the volume of the sorbent material as well as the scaffolding to which it is adhered, which can vary from metallic foams to gel composites. Water-harvesting efficiency (η_{AWH}) of the system is a device-level metric that aims to quantify the capabilities of the system with respect to ideal, regardless of material choice. As a mass-based efficiency measurement, it weighs the actual water production of the system with the theoretical output of the sorbent material, given a specific amount of material, uptake swing, and number of cycles per day of sorption. This denominator equates to the maximum potential for water output from a given sorbent material in a specific environmental condition, cycled according to device operation. For instance, a low water-harvesting efficiency states that the device is not ideally cycling a given material or suffers from system-level non-idealities such as vapor leakage, water retention, or excessive thermal mass. Maximizing the water-harvesting efficiency results in a device that is capable of ideally utilizing a given material, thus minimizing leakage, heat loss, and ramp-up times and operating with ideal airflow vapor delivery and power consumption.

These four metrics can readily be applied to a variety of SAWH devices, both passive and active, for a variety of heating

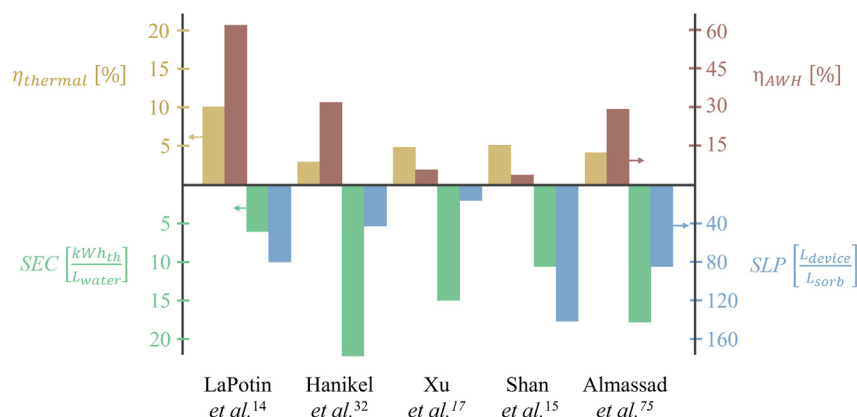


Figure 7. Performance metrics for SAWH

Evaluation of proposed SAWH performance metrics for five example devices, based on published data. Efficiencies $\eta_{thermal}$ and η_{AWH} are targets to be maximized, while SEC and SLP are minimization metrics.

source options and system vapor flow configurations. Figure 7 shows these four performance metrics evaluated for five SAWH devices, based on available published data, with detailed calculations provided in the supplemental information. Critically, these metrics are dependent on environmental conditions and thus each device has a range of values for each metric. For our analysis we considered testing conditions as close to 25°C and 30% RH as possible based on reporting (room temperature, arid climate); however, a precise comparison between devices should be avoided due to slight variations in testing conditions. In future work, there is the potential to report each performance metric over a broad range of environmental conditions, emphasizing ideal versus non-ideal device operation, and allowing for comparison across technologies. As shown in Figure 7, single-cycle passive devices exhibit significantly higher η_{AWH} efficiencies compared with multicycle systems, highlighting the potential to improve the use of a given sorbent material in a multicycle system through device optimization. By considering all aspects of a SAWH device, including sorbent bed, heating source, condenser, and airflow delivery, future devices can potentially maximize the water output of a given sorbent material and drastically increase the performance of SAWH devices.

CONCLUSIONS AND FUTURE OUTLOOK

SAWH technology has the potential to address the current global water crisis, especially in arid regions; however, current device-level limitations inhibit the deployment of truly autonomous actively driven devices capable of high daily water output (>2 L/day per person) to meet daily human needs. While sorbent material innovations have drastically increased water sorption capabilities, efficient control of the adsorbate through the adsorption, desorption, and condensation processes remains a limiting factor for effective SAWH devices. By considering the system as a whole, consisting of the critical subcomponents sorbent bed, heating source, condenser, and airflow delivery, there is the potential to produce an optimal device in terms of chosen performance metrics and dependent, subcomponent-defined design variables. Target performance metrics vary based on target application, but should include the need for compactness

(SLP), minimal energy consumption (SEC), maximal thermal efficiency ($\eta_{thermal}$), or maximal performance of a given sorbent material (η_{AWH}). A key contribution to the improvement of SAWH devices is the consideration of airflow delivery and impact on performance, including vapor recovery ratio and η_{AWH} . For a multicycle system, airflow delivery dictates not only

a portion of power consumption but also water vapor loadings in the sorbent bed and cycle times. Furthermore, an essential improvement to operation will be the actuation and sealing of future SAWH devices, minimizing both user input and water vapor leakage during the desorption and condensation process. In conclusion, SAWH is an attractive solution for generating potable water in arid regions; however, system challenges must be rapidly addressed to realize functional devices capable of efficiently producing water at the liter scale. Here, we summarize previous specific device achievements and generally decompose SAWH devices into four critical subcomponents, discussing how researchers may consider the coupling between them to produce an optimal device. In addition, we define four standardized performance metrics, which may be utilized to evaluate and improve a SAWH design. To complete the design considerations for the next generation of SAWH devices, a comprehensive cost analysis should be performed. Generally, SAWH requires the integration of sorbents, which can be costly to develop or more often modify by impregnating with salts such as LiCl, CaCl₂, MgCl₂, etc. In addition, the design and engineering of the system, including the incorporation of sorbent regeneration processes and water collection mechanisms, adds to the overall cost. The cost of water produced by SAWH systems can potentially be competitive compared with the cost of desalinated water or bottled water, especially in areas with high humidity and limited access to freshwater sources. Furthermore, SAWH systems can harness atmospheric moisture, providing a local and sustainable water supply without the need for extensive infrastructure. Finally, SAWH systems offer environmental benefits by reducing the reliance on desalination plants, which consume large amounts of energy and contribute to greenhouse gas emissions. In addition, they help minimize plastic waste associated with bottled water production and transportation. By looking at the technology holistically and designing systems for cost-effective scale-up, SAWH may begin to aid in potable water procurement for the growing water-stressed regions of the globe.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.device.2023.100052>.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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